
CS771 Minor Assignment 3

Diya Patel diyahp24@iitk.ac.in
Saurya Singh sauryas24@iitk.ac.in
Medini Mudraboina medinim24@iitk.ac.in
Shruti Sahu shrutisahu24@iitk.ac.in
Vasu Goyal vasug24@iitk.ac.in
Deepak Kumar deepakk24@iitk.ac.in

Question 1

We implemented the EM algorithm for mixed regression using $n_components = 4$. To evaluate the impact of the number of iterations on model performance, we varied the n_iter parameter across 10 values: $\{1, 2, 3, 5, 10, 15, 20, 30, 40, 50, 80, 100\}$. For each execution, we trained an auxiliary classification model to predict the best regression model for the unseen test data to calculate a realistic Test MAE.

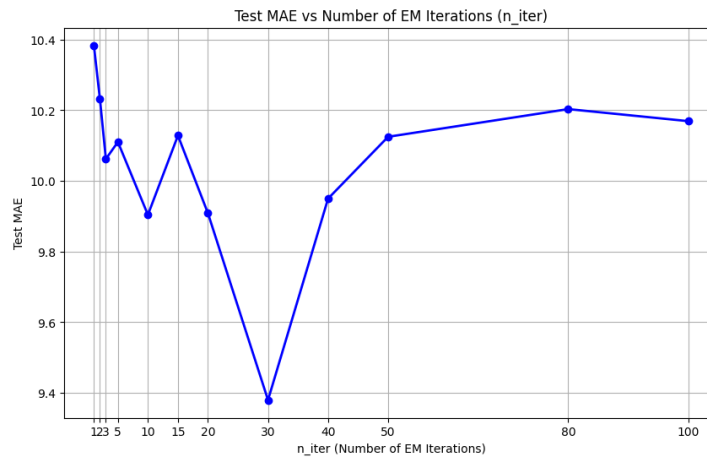


Figure 1: Test MAE vs. Number of EM Iterations (n_iter) with 4 components.

Discussion: As observed in the plot, the Test MAE exhibits significant volatility during the early stages of training. While there is a sharp, temporary drop in error at $n_iter = 30$, the performance continues to fluctuate before finally stabilizing between 40 and 50 iterations. Therefore, we can identify $n_iter = 50$ as a reasonable knee point. Beyond this threshold, increasing the number of iterations (e.g., towards 80 or 100) provides little to no stable improvement. This plateau indicates that the EM algorithm has effectively converged and additional iterations are unnecessary.

Question 2

In this section, we explore how changing the number of mixture components ($n_components$) impacts the model's test Mean Absolute Error (MAE). Building on our findings from Part 1, we kept the number of EM iterations fixed at 50 ($n_iter = 50$) to ensure a stable baseline.

We tested the model across several configurations, specifically setting the number of components to $n_components \in \{1, 2, 4, 8, 16, 32\}$. For each setup, we used the Expectation-Maximization

algorithm to fit the regression models. Afterward, we trained an auxiliary classification model to identify the best component for the unseen test data.

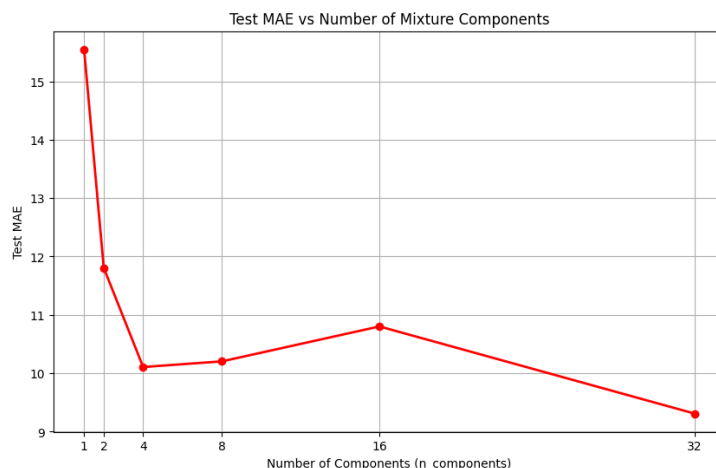


Figure 2: Test MAE vs. Number of Mixture Components ($n_{components}$), with EM iterations fixed at 30.

As Figure 2 illustrates, adding more components initially leads to a sharp drop in the test MAE, as the model becomes better at distinguishing between different pollution patterns. However, pushing the number of components too high (such as 16 or 32) causes the performance to plateau or even worsen. This happens because the training data gets spread too thin across the many components. As a result, the individual regression models don't have enough data to generalize well, and the auxiliary classification task becomes unnecessarily complex.

Question 3

For $n_{components} = 1$, Trivially one component — covers all data. Nothing to compare.

For $n_{components} = 2$, the EM algorithm splits data into two groups driven by the diurnal ozone cycle: a nighttime cluster (low O3 ~ 14 ppb, mean hour ~ 3 AM, cooler temperature) and a daytime cluster (high O3 ~ 55 ppb, mean hour ~ 1 PM, warmer). The dominant distinguishing features are Time and o3op1 (mV), both of which track the photochemical buildup of ozone during daylight hours.

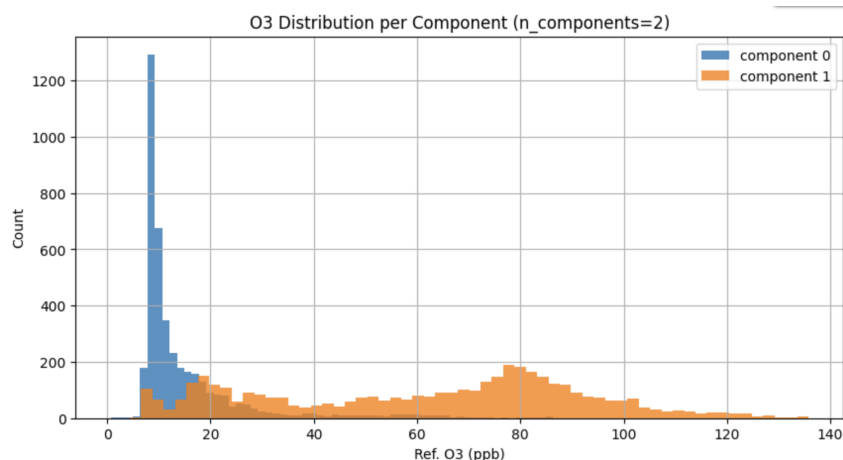


Figure 3: Density distribution of Original Ref. O3(ppb) for $n_{components} = 2$. The model acts as a broad binary threshold, splitting the data into normal baseline air quality and elevated pollution levels.

For $n_components = 4$ (the lecture default), the four components correspond to four O3 intensity bands mapping to late-night, morning/evening transition, afternoon, and peak-ozone periods. This matches the four-component interpretation noted in the lecture code. Temperature increases monotonically across components but is a consequence rather than a cause — the underlying driver remains the time-of-day ozone photochemistry.

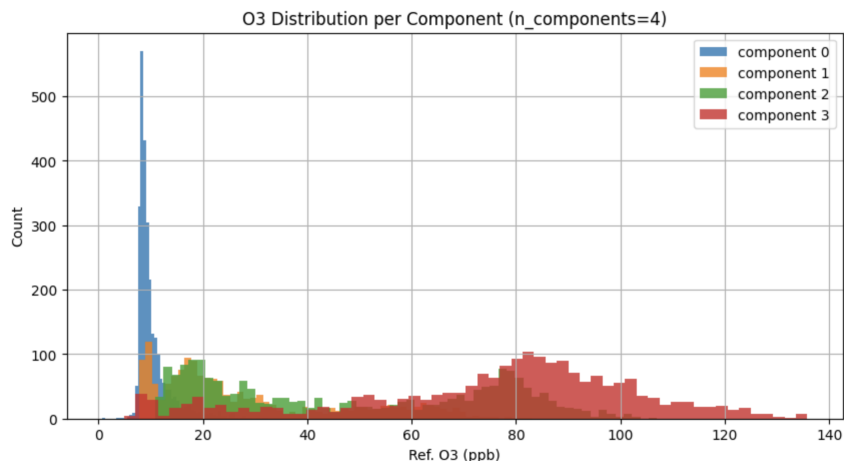


Figure 4: Density distribution of Original Ref. O3(ppb) for $n_components = 4$. The model identifies four distinct O3 regimes: very low (component 0, nighttime), low-moderate (component 1, morning/evening), moderate-high (component 2, afternoon), and peak O3 (component 3, mid-afternoon). This split aligns naturally with the diurnal ozone cycle and gives the most physically meaningful component separation.

For $n_components = 8$, the components become difficult to interpret. Adjacent components (e.g. C0 and C1) share nearly identical O3 ranges and hour-of-day distributions, differing only in subtle sensor voltage variations. Data is spread too thinly ($\sim 12\%$ per component), individual regression models generalize poorly, and the auxiliary classifier cannot reliably distinguish 8 near-identical clusters — hence test MAE does not meaningfully improve.

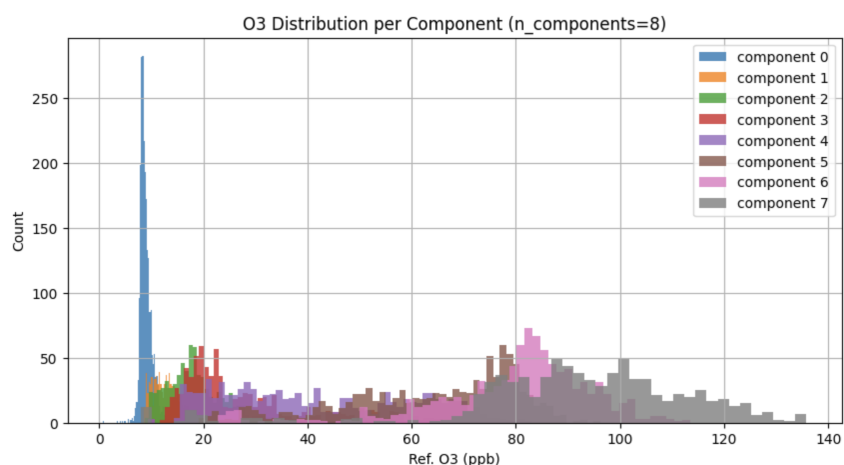


Figure 5: Density distribution of Original Ref. O3(ppb) for $n_components = 8$. The model over-partitions the data into eight heavily overlapping components, especially in the 0–40 ppb range. Adjacent components are physically indistinguishable, as splits are driven by subtle sensor voltage variations rather than meaningful pollution regimes. Training data spreads too thinly, offering no significant MAE improvement over $n_components = 4$.

The key takeaway for Part 3: Time and o3op1(mV) are the primary separators — they track the diurnal ozone photochemistry cycle. At `n_components = 2` and `4` the components are physically meaningful (time-of-day O3 bands); at `n_components = 8` the split becomes noise-driven.

Question 4

We know that if removing a feature causes the Test MAE to spike dramatically, that feature is highly important and if removing a feature barely changes the Test MAE compared to your baseline, that feature contributes very little.

To determine feature contribution, we analyze the ablation study graphs. The baseline model (all features included) achieves a Test MAE of approximately 9.3 to 10.1 (depending on `n_components`). When a feature is excluded, any resulting increase in Test MAE represents a drop in performance. The feature that causes the largest performance drop when removed is the most critical to the model.

Excluding Time: The **Figure 3** graph shows a moderate impact. The error rises to about 11.9 at `n_components=4`. The model misses the diurnal (time-of-day) context, but can still somewhat infer it from the temperature and sensor readings.

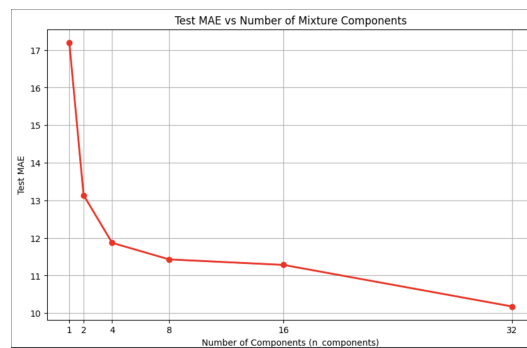


Figure 6: Feature Ablation Study: Impact of Excluding 'Time'

Excluding Temp (C): The **Figure 4** graph shows a similarly harsh penalty. At `n_components=4`, the error spikes to roughly 13.7. Temperature is clearly a vital physical parameter driving the photochemical reactions that the model is trying to capture.

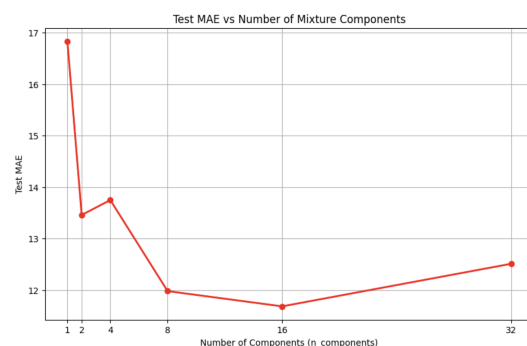


Figure 7: Feature Ablation Study: Impact of Excluding 'Temp(C)'

Excluding RH (%): Looking at the **Figure 5** graph, the Test MAE only rises slightly (about 11.3 at `n_components=4`) and drops back down to ~ 10.0 at `n_components=16`. This curve stays the closest to your original baseline performance.

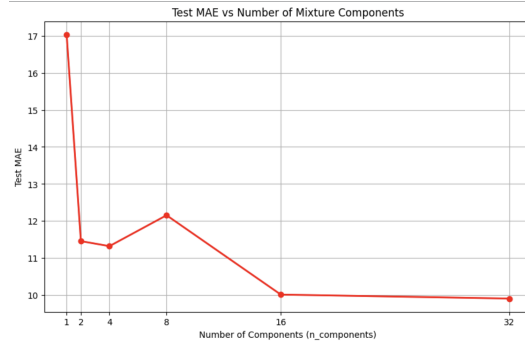


Figure 8: Feature Ablation Study: Impact of Excluding 'RH'

Excluding o3op1 (mV): Looking at the **Figure 6** graph, removing this sensor reading devastates the model. The Test MAE stays extremely high (around 13.6 at **n_components=4** and remains above 12.5 even at higher components). This massive drop in accuracy indicates the model relies heavily on this raw sensor voltage to make predictions.

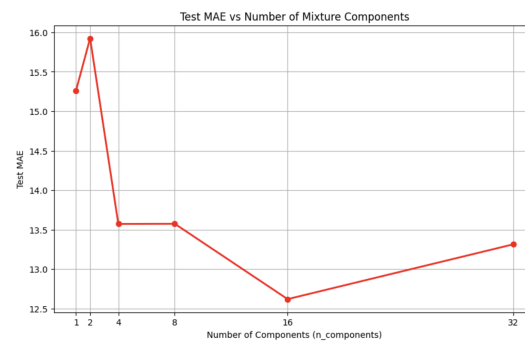


Figure 9: Feature Ablation Study: Impact of Excluding 'O3op1 (mV)'

Excluding o3op2 (mV): The **Figure 7** graph shows a moderate-to-high penalty. The error jumps to about 12.7 at **n_components=4**. It is important, but perhaps slightly less critical than **o3op1**.

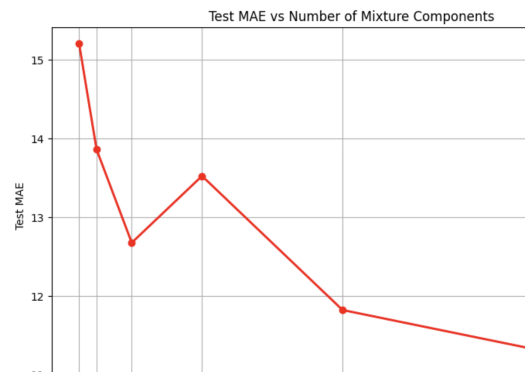


Figure 10: Feature Ablation Study: Impact of Excluding 'O3op2 (mV)'.

- **Feature that contributes the MOST: o3op1 (mV) (and Temp (C)).** Excluding either of these causes the most drastic loss in predictive accuracy, pushing the Test MAE significantly higher than the baseline. The model cannot function accurately without them.
- **Feature that contributes the LEAST: RH (%).** Excluding Relative Humidity results in an error curve that remains very close to the baseline performance. The model can accurately

predict pollution levels even when it is completely "blind" to the humidity, making it the least valuable feature in this dataset.

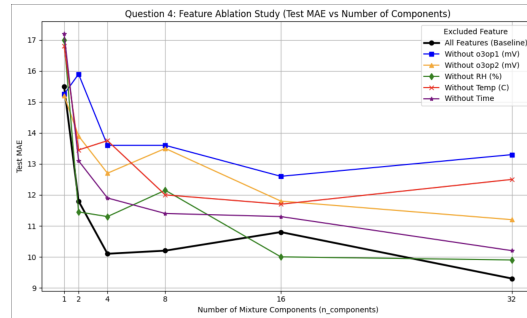


Figure 11: This is the combination of all the asked combination of features .

Question 5

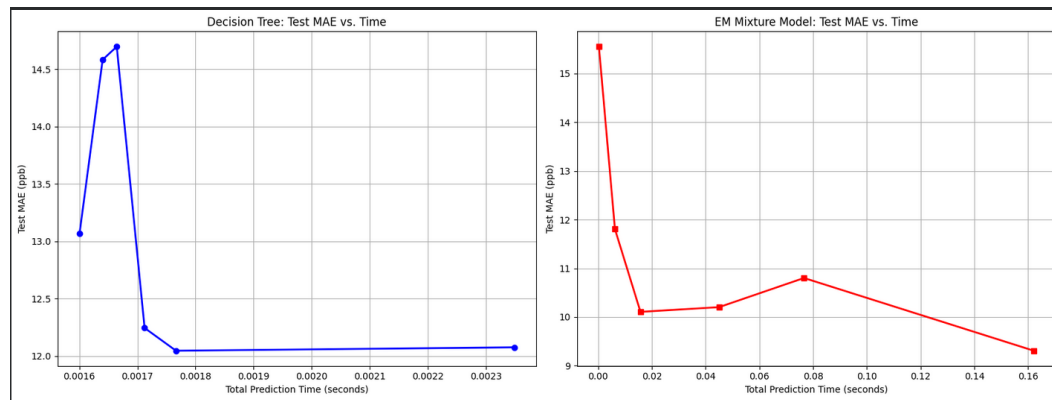


Figure 12: Performance on MAE and test time comparison of Decision Tree model and EM Algorithm

For the DecisionTreeRegressor, after performing hyperparameter tuning, we determined that setting the criterion to 'absolute_error' and the splitter to 'best' yields the optimal performance across the various depths tested.

As can be seen in Figure 12 , comparing the Decision Tree Regressor to the EM Mixture Model shows that there is a heavy trade off between speed and accuracy.

Speed: The Decision Tree is faster. Its total prediction time over the entire test set ranges from approximately 0.0016 to 0.0023 seconds. In contrast, the EM Mixture Model takes significantly longer time with prediction time going up to 0.16 seconds as number of mixture components increases.

This is because, while a decision tree only does simple binary threshold checks at every node, the mixture model involves passing data through a classification model, and then evaluating many regression equations.

Accuracy: The EM Mixture Model is more accurate. The Test MAE of the Decision Tree tends to 12 while the EM Mixture Model achieves an MAE of around 10.1 even at lower number of components and when number of components are increased, it decreases further to around 9.3.

This is because, while a decision tree predicts a constant, average value for all data points that fall into a specific leaf, the EM Mixture Model fits distinct linear regression equations within each component. Because AQI features have continuous, proportional relationships with pollution levels, the EM Mixture Model is able to make accurate predictions.